Ashraf Muumin

Alabama • chiriashraf@gmail.com • https://chirix.netlify.app • (442) 224-9933

Education

The University of Alabama — Tuscaloosa, AL
Doctor of Philosophy, Computer Science — Aug 2026 – Present
Research Focus: Neurosymbolic AI, Physical AI, ML Systems, Agentic AI Systems

Illinois Institute of Technology — Chicago, IL
Master's, Information Technology (Software Development). GPA 4.0 — May 2026
Relevant Coursework: Application Development Methodologies, Advanced Software Development

Experience

The University of Alabama — Tuscaloosa, AL
Graduate Research Assistant, Neurosymbolic AI — Aug 2026 – Present

- **Engineered and published** a simulation-based dataset of **11,105 physical-reasoning questions across 6,340 scenarios**, using Python, to cover robotic manipulation, vehicle dynamics, object interactions, and suspended loads.
- Developed an **LLM annotation and validation pipeline** producing **596 enriched QA records across five task families**, combining schema-constrained generation, evidence traceability, automated numerical and temporal checks, and a three-reviewer quality audit.
- Built a reproducible **GPU evaluation framework for five LLMs across nine configurations and two ablations**, scoring **2,200 evaluations** with typed metrics, failure analysis, and bootstrap confidence intervals; packaged datasets, prompts, and scoring tools for research release.
- Developing an **LLM-based regulatory document review system** using public ICH and FDA standards, in collaboration with an industry partner.

Lumena Energy — Chicago, IL
Full Stack Software Engineer — Sep 2025 – Dec 2025

- Deployed an **on-prem proprietary AI model** for internal and customer-facing use, optimizing inference performance and secure model access.
- Managed **CI/CD pipelines** for automated build, test, and deployment workflows targeting **GCP**, increasing deployment reliability and speed.
- Containerized production services with **Docker**, enabling predictable deployments and environment consistency.

AIFS — Chicago, IL
.NET Backend AI Engineer — Jun 2025 – Aug 2025

- Architected and deployed scalable **Azure Function Apps** in **.NET** that supported over 1,000 requests/sec, contributing to cost-efficient cloud-based infrastructure aligned with enterprise performance needs.
- Built **agentic orchestration workflows** using **Semantic Kernel** and **Microsoft's Agentic Framework**, enabling parallel task execution and intelligent automation capabilities.
- Enhanced **Python data ingestion and transformation pipelines** by integrating **Azure Durable Functions**, increasing reliability and scalability of long-running workflows.

Skills

- **Technical**: Python, JavaScript, TypeScript, C#, .NET, SQL, Bash, NextJS, NumPy, Pandas, Git, Figma, Azure, AWS
- **Competencies:** Neurosymbolic AI, LLM Evaluation, Agentic AI, Experimental Design, Agile, RESTful APIs, Microservices, SDLC, User Interface, User Experience, Prototyping, Responsive Design